

SZL Holdings — Operational Briefing (v2)

Prepared for Empire APEX Accelerator

Refresh: May 5, 2026 — supersedes v1 (May 3, 2026)

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1. What changed since the May 3 briefing

The May 3 briefing introduced SZL Holdings as a unified governed-intelligence platform with seven product surfaces, an open-source runtime, and a DOI-pinned thesis. In the 48 hours since that briefing was assembled, the following material updates landed and are reflected in this document:

1. **A second, peer-style paper was published** — the Ouroboros Thesis v9 (*The Lutar Invariant Family $v1 \rightarrow v7 \rightarrow \Omega$: From Three-Term Foundation to Bianchi-Closed Fiber Bundle*) and a companion v10 audit paper (*The Audit Closure Operator Λ_{10}*) were deposited to GitHub on May 5 with PDF artifacts attached to the GitHub release source archives so the Zenodo webhook re-fires with each tag. v9 is 17 pages, v10 is 11 pages plus an Appendix A essay and a Lutar-family one-pager.
2. **A11oy gained four new operator surfaces** — Argo (experience-era decision engine with champion policies, mirror eval, counterfactuals, and reward-hacking guardrails), PSYCHE (emergent-sentience observatory), the Trust Center / Trust Exchange / Public Trust Portal triad, and Agent Zero Trust.
3. **A new vertical surface, ROSIE — Unified Decision Fabric**, was promoted from internal tooling to a registered customer-facing artifact. There are now eight customer-facing surfaces, not seven.
4. **The Covenant Proof Standard (CPS) shipped as a first-class API** — `/api/cps/payloads`, `/api/cps/runs`, with three flagship payloads, tiered approval, rollback, and a maturity-mode gate.
5. **A live agent gateway service** (`artifacts/api-server: agent-gateway`) now sits in front of every agent action and enforces OPA bundle policy at the runtime boundary.

6. **A documented investor demo path** (docs/audits/INVESTOR_DEMO_PATH.md) and per-artifact audit reports for all eight surfaces were committed, so the platform can be walked end-to-end on a known-good route.
7. **Surgical Series-A polish** was applied across all seven prior verticals — placeholder URLs purged, demo-mode copy scrubbed, broken links removed, date-rotted fields rewired to live timestamps, fabricated example data deleted.

The platform's core thesis has not changed: governed AI cognition, proof-chain by construction, math written down. What changed is the operational evidence backing that thesis.

2. The Governed Command Layer (unchanged)

The architectural model remains: five stages, run as one continuous loop.

1. **Ingest** — Signals normalized across operational, security, financial, document, sensor, and third-party systems.
2. **Score** — Risks and opportunities scored in real time against policy.
3. **Decide** — Decisions generated with AI support, gated by policy where the policy says they must be gated.
4. **Act** — Actions executed with full traceability — every call recorded, every input hashed, every approver named.
5. **Verify** — Outcomes verified against intent and recorded into auditable history that can be replayed.

The convergence properties of this loop (when it terminates, what its proof-chain has to look like, and the closed-form Λ bound that makes the bound auditable) are now formalized in the Ouroboros Thesis paper family v1 $\rightarrow$ v10 (Section 7).

3. Platform Surface — Live Today (refreshed)

The platform now runs as **eight customer-facing product surfaces** orchestrated by a single decision-intelligence layer (A11oy).

A11oy — Orchestration + Decision Intelligence + Trust Plane

Slug: /a11oy/

A11oy is the orchestration control plane and the decision-intelligence surface. It registers agents, applies validator policy, gates approvals, and is where operators monitor and govern bounded-loop runs across every vertical. Since the May 3 briefing, A11oy has gained:

- **Argo** (/a1loy/argo , /argo-bridge) — an experience-era decision engine. Champion policies (6 live), self-play arena with replay playback, mirror evaluation, counterfactual rollouts, reward-hacking guardrails, world-model accuracy at 89.1%, throughput 31.4 ev/s.
- **PSYCHE** — emergent-sentience observatory; tracks behavioral signals and self-modeling metrics across the agent fleet.
- **Trust Center** (/trust-center) — the constitutional surface; proof, covenants, and attestation in one place.
- **Trust Exchange + Public Trust Portal** (/trust-exchange , /public-trust-portal) — the externally-facing proof-distribution surface.
- **Agent Zero Trust** (/agent-zero-trust) — runtime policy gate for every agent action; backed by the live agent-gateway service.
- A live Claude Sonnet 4.6 advisor at /a1loy/chat (introduced in v1 of this dossier; still operational).

Sentra — Cyber Resilience Command (codename TENAX)

Slug: /sentra/

Continuous threat-posture monitoring, evidence packs, risk-tier escalation gates wired into the runtime contract. Since May 3, Sentra has shipped a **governed adversary loop** (/sentra/governed-adversary-loop) that emits a six-step proof chain through A11oy, plus a KORA cyber-resilience trust feed that publishes threat posture into the Trust Plane.

Amaru — Convergent Reverse-ETL (codename Conduit)

Slug: /conduit/

Append-only delta log, hash-verified ingest, three-witness reconciliation primitive (Frustum) from the Ouroboros Thesis. Since May 3, Amaru shipped **ten original innovations beyond the GitHub field of competing reverse-ETL tools** (catalogued under the “Conduit one-of-one” engineering pass). The dashboard throughput chart was rewired from a hard-coded clock to `Date.now()` so the rolling window never freezes.

Terra — Real Estate Intelligence (codename DOMAINE)

Slug: /terra/

Distressed-property discovery, ownership analysis, pipeline management, deal execution. NYC and NYS data live. Since May 3: synthetic confidence/escalation columns were removed from the property-detail tenant table; banner stacking on narrow viewports fixed.

Vessels — Maritime Intelligence (codename SEXTANT)

Slug: /vessels/

Positions, voyage economics, compliance, exception management. Sanctions screening and dark-vessel detection. Since May 3: footer cleaned of personal links; “Updated 12s ago” date-rot replaced with a live AIS-feed indicator.

Counsel — Legal Matter Command

Slug: `/counsel/`

Matters, obligations, and legal exposure surfaced as a command surface. Policy-gated human review, evidence-bound recommendation, citation-verified output. Since May 3: a synthetic “SEC Filing Deadline” injection was removed; the empty-state now renders honestly when there is no real data.

Carlota Jo — Concierge Advisory Operations

Slug: `/carlota-jo/`

Premium service brand for individuals who require precision and discretion. Proof-Chain decision-receipts bound to the runtime. Since May 3: full **competitive-intel and ML-forecast module**, deeper A11oy mesh integration, command-palette navigation rewritten to be `BASE_URL`-aware across all 23 commands; invented “847 enterprise contracts” sample exchange removed; intake `timeline: TBD` default replaced with honest copy.

ROSIE — Unified Decision Fabric (NEW since May 3)

Slug: `/rosie/`

Promoted to a registered customer-facing artifact in the May 4–5 cycle. Six operator pages: Identity, Optimizer, Fabric, Research, Proof, Evidence Bench. ROSIE is the integration surface for cross-vertical decision payloads — it consumes the CPS standard (Section 5) and renders the proof bench against live runs.

4. The Six Platform Primitives (unchanged)

Primitive	What it does
Outcome Graph	Every action is a node; every cause is an edge. The graph is the system of record for “what did we actually decide and why.”
Proof Chain	Every output carries a hash chain back to the inputs and the human who approved each gated step.
Covenant Policy	Declarative policy that decides which actions are auto-allowed, which require human approval, and which are forbidden.
Decision Simulation	Decisions can be replayed against historical state to test policy changes before they ship.
Workflow Engine	The bounded-loop scheduler that runs the five stages of the governed command layer.
Event Fabric (PRISM Bus)	The append-only event spine. Everything that happens publishes here.

5. Covenant Proof Standard (CPS) — new in v2

CPS is a payload-and-receipt standard the platform now exposes as a first-class API. Every cross-vertical workflow can be packaged as a CPS payload, executed against a tenant, and resolved through a tiered approval ladder with auditable rollback.

Endpoint	Behavior
GET /api/cps/payloads	List registered payloads
GET /api/cps/payloads/:id	Resolve a payload definition
POST /api/cps/runs	Execute a payload run
GET /api/cps/runs/:id	Inspect a run, its proof receipts, and approval state
POST /api/cps/runs/:id/approve	Approve a gated step at the caller’s tier
POST /api/cps/runs/:id/rollback	Roll a completed run back to a prior verified state
POST /api/cps/payloads/:id/maturity	Promote/demote a payload’s maturity mode

Three flagship payloads ship with the standard. They cover the cross-domain shape investors and procurement officers care about most: incident → governed action → audit close.

The CPS layer is the reason ROSIE exists as its own artifact: ROSIE is the operator surface for CPS payloads, and the pattern repeats per-lane (Vessels, Terra, Counsel, Carlota Jo) as those

payloads come online.

6. Verified Platform Numbers (re-verified May 5, 2026)

Every number below was produced by re-running the documented verification command in `SOURCE_OF_TRUTH.md` . No estimation.

Metric	Verified value	Δ vs May 3
Registered artifacts (artifact.toml files)	9	unchanged
Customer-facing product surfaces (live)	8 + the A11oy orchestration layer	+1 (ROSIE)
Database tables (live, provisioned)	848	unchanged
API endpoint declarations	5,524	unchanged
Industry verticals	7	unchanged
Monorepo packages (<code>packages/</code> + <code>lib/</code>)	126	unchanged
DB schema files	170	unchanged
CI workflows	23	unchanged
Declared environment variables	213	unchanged
Platform primitives	6	unchanged
RBAC roles	11	unchanged
Ouroboros runtime test calls	133	unchanged
Codex-kernel test calls	29	unchanged
Ouroboros packages (<code>@workspace/ouroboros-*</code>)	28	new disclosure
Ouroboros guardrails tests passing	62	new disclosure
Formal axes in the Lutar invariant family	9	new disclosure
Codex v11 nodes / typed edges	76 / 95 across 11 domains	new disclosure

Numbers will continue to drift as the platform grows. The discipline is that they are re-verified by command, not estimated. This is a contracting hygiene point, not a marketing one.

7. Public Proof — refreshed

Three artifacts are public on GitHub today, all under `szl-holdings` , all with verifiable history:

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Ouroboros Runtime (open source)

- Repository: `https://github.com/szl-holdings/ouroboros`
- Latest release: **v6.2.0** — “v4 validator FUNCTIONS (runnable)”
- Tests, dependabot, CodeQL, [SECURITY.md](#).

Ouroboros Thesis (DOI-pinned, multi-paper)

Repository: `https://github.com/szl-holdings/ouroboros-thesis`

Paper	Title	Pages	Released	DOI
v3-2.0.0	The Loop Is the Product: Measuring Bounded Recursion as a System Primitive for Auditable AI	—	2026-05-02	<code>10.5281/zenodo.19944926</code>
v9-1.0.0	The Lutar Invariant Family v1 → v7 → Ω: From Three-Term Foundation to Bianchi-Closed Fiber Bundle	17	2026-05-05	minted via Zenodo webhook on tag re-fire
v10-1.0.0	The Audit Closure Operator Λ_{10} : Formalising the Implementation Contract of the Lutar Family	11 + Appendix A essay + Lutar-family one-pager	2026-05-05	minted via Zenodo webhook on tag re-fire

License: CC-BY-4.0 (academic distribution). ORCID: `0009-0001-0110-4173` .

The May 2 release was an audit-supported rewrite that explicitly retracted a prior version after a self-audit identified residual fabrications in announcement materials. The $v9 \rightarrow v10$ sequence is the next step in that discipline: v10’s only purpose is to audit v9’s implementation chain and certify, layer by layer, that every formula in v9 actually executes against the live shipping repo. **No new physical L-term is introduced in v10.** It is a meta-invariant on v9.

Live LaaS API (Lutar-as-a-Service)

`POST /api/ouroboros/lutar/v10` runs the v10 audit closure operator against the live shipping repo on every test run; result is a typed object backing the `lutar_v10` codex node. This is the operational binding from paper to production.

8. Government Contracting Alignment (refreshed)

The work appears to align with five areas of increasing federal and state demand. These are alignment statements, not claims of past performance.

Operational transparency and auditability

The Outcome Graph and Proof Chain primitives answer the question every IG, every audit committee, and every Congressional oversight letter eventually asks: “show your work.” CPS makes the answer interoperable across systems.

Cybersecurity and risk management

Sentra is the customer-facing surface. The deeper architectural fit is that the runtime treats every action as a security event by default — logged, signed, replayable. The new governed adversary loop demonstrates this end-to-end across Sentra ↔ Alloy in a six-step proof chain.

AI governance and responsible automation

Covenant Policy gates AI outputs. The new live agent gateway enforces OPA policy at the runtime boundary; Argo’s reward-hacking guardrails and mirror eval close the loop on “is the policy actually working.” Relevant to OMB M-24-10 and the NIST AI RMF.

Cross-system interoperability and data lineage

Amaru (convergent reverse-ETL) plus the three-witness reconciliation primitive (Frustum) gives a defensible answer to “single trusted view of state across N systems.” The ten new innovations from the Conduit one-of-one pass extend this beyond what the open-source field offers today.

Auditable AI proof distribution

The Trust Plane (Trust Center, Trust Exchange, Public Trust Portal) is the externally-facing surface for distributing proof packets to regulators, auditors, and partner systems. CPS payloads + Trust Plane = a procurement-grade distribution contract for proof-of-decision evidence.

9. What SZL Holdings Is Not Claiming (unchanged)

- SZL Holdings is **not currently performing on any federal contract**.
- SZL Holdings has **not received a federal cloud authorization**, an ATO, or any DoD impact-level designation.
- SZL Holdings has **not been audited by an outside firm**.
- SZL Holdings has **no signed customer contracts** for the platform itself at this writing.
- SZL Holdings is **a single-founder operation** (Stephen Lutar). There is no team to misrepresent.
- The numbers in section 6 are **internal platform metrics** — they describe what has been built, not revenue or users.

The strength of the position is the public proof and the verifiable runtime, not pretended traction.

10. Suggested Discussion for May 6 (unchanged priorities)

In priority order:

1. **NYS pathway.** OGS centralized contract? On-call advisory pool? Subcontract / mentor-protégé under a NYS prime?
 2. **Federal pathway.** SBIR Phase I as the fastest credible entry point? GSA MAS via a sponsoring contract holder? Sources-sought response in AI governance / continuous-monitoring?
 3. **Registration sequence.** [SAM.gov](https://sam.gov) UEI, CAGE, NAICS selection, small-business size standard, set-aside eligibility review, NYS Vendor Responsibility Questionnaire — order and realistic timelines.
 4. **Capability statement format.** What NYS and federal procurement officers actually want on the page.
 5. **Realistic targeting.** Which agencies actually buy this kind of work, and which buy it from sole-founder firms.
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11. Documents Attached

- This briefing (`dossier/v2/APEX_v2_Operational_Briefing.md`).
 - Investor pitch deck (`dossier/v2/APEX_v2_Investor_Pitch_Deck.md`).
 - Business proposal (`dossier/v2/APEX_v2_Business_Proposal.md`).
 - Meeting script (`dossier/v2/APEX_v2_Meeting_Script.md`).
 - Live demo guide (`dossier/v2/APEX_v2_Live_Demo_Guide.md`).
 - Capability statement (`dossier/v2/APEX_v2_Capability_Statement.md`).
 - Audit delta May 4 → May 5 (`dossier/v2/APEX_v2_Audit_Delta_May5.md`).
 - Dossier index (`dossier/v2/APEX_v2_DOSSIER_INDEX.md`).
 - Live screenshots of each product surface (`dossier/screenshots/`).
 - Ouroboros Thesis v3 / v9 / v10 PDFs (Zenodo + GitHub Releases).
 - `SOURCE_OF_TRUTH.md` and `audit/source-of-truth.json` audit trail in the platform repository.
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12. Closing

The premise has not changed. Governed AI cognition will be a procurement requirement before long, not a procurement preference. Two days of additional shipping made the proof denser: a second peer-style paper, a third (the audit closure paper), a CPS standard, a live agent gateway, four new Trust-Plane surfaces, an eighth vertical (ROSIE), and a documented investor demo path that runs the proof spine end-to-end.

The May 6 conversation remains about how to get this in front of buyers responsibly.

Thank you for the time.

— Stephen Lutar Founder, SZL Holdings stephenlutar2@gmail.com